projects

```
PK id: VARCHAR(32)
name: VARCHAR(120)
workflow_key: VARCHAR(60)
description: TEXT
created_at: DATETIME
```

runs

```
PK id: VARCHAR(32)
FK project_id: VARCHAR(32)
workflow_key: VARCHAR(60)
status: VARCHAR(20)
llm_mode: VARCHAR(20)
model: VARCHAR(80)
seed: INTEGER
fault_type: VARCHAR(40)?
task_input: JSON
trace: JSON?
trace_sha256: VARCHAR(64)?
error: TEXT
created_at: DATETIME
finished_at: DATETIME?
```

check_results

```
PK id: INTEGER
FK run_id: VARCHAR(32)
check_id: VARCHAR(40)
check_version: INTEGER
scope: VARCHAR(10)
step: VARCHAR(40)
type: VARCHAR(20)
severity: VARCHAR(10)
status: VARCHAR(10)
observed: JSON?
expected: JSON?
message: TEXT
computation: JSON
```

Legend

- investigation data
- housekeeping
- evaluation ground truth
- foreign key (bar = one, fork = many)
- reference by id, no constraint

Design choices

The trace of a run is one JSON column (runs.trace); steps and edges are derived from it on read. evidence and audit_log have no update or delete route, and refer to other rows by id only, so they outlive what they describe.
? marks a nullable column.

jobs

```
PK id: VARCHAR(32)
kind: VARCHAR(20)
target_id: VARCHAR(32)
status: VARCHAR(15)
progress_done: INTEGER
progress_total: INTEGER
message: TEXT
created_at: DATETIME
started_at: DATETIME?
finished_at: DATETIME?
```

fault_injections

```
PK run_id: VARCHAR(32)
fault_type: VARCHAR(40)
target_step: VARCHAR(40)
field_path: VARCHAR(120)
original_value: JSON
injected_value: JSON
note: TEXT
```

findings

```
PK id: VARCHAR(32)
FK run_id: VARCHAR(32)
kind: VARCHAR(20)
severity: VARCHAR(10)
title: TEXT
detail: TEXT
check_id: VARCHAR(40)?
symptom_step: VARCHAR(40)
status: VARCHAR(20)
created_at: DATETIME
updated_at: DATETIME
```

hypotheses

```
PK id: VARCHAR(32)
FK finding_id: VARCHAR(32)
rank: INTEGER
step_id: VARCHAR(40)
agent: VARCHAR(80)
method: VARCHAR(20)
evidence_kind: VARCHAR(30)
score: FLOAT
rationale: TEXT
evidence: JSON
status: VARCHAR(20)
```

experiments

```
PK id: VARCHAR(32)
FK hypothesis_id: VARCHAR(32)
finding_id: VARCHAR(32)
status: VARCHAR(20)
llm_mode: VARCHAR(20)
n: INTEGER
m: INTEGER
base_seed: INTEGER
rule_version: VARCHAR(10)
intervention: JSON?
summary: JSON?
verdict: VARCHAR(20)?
error: TEXT
created_at: DATETIME
finished_at: DATETIME?
```

audit_log

```
PK id: INTEGER
ts: DATETIME
actor: VARCHAR(80)
action: VARCHAR(40)
entity_type: VARCHAR(20)
entity_id: VARCHAR(32)
detail: JSON
```

reviews

```
PK id: VARCHAR(32)
FK finding_id: VARCHAR(32)
reviewer: VARCHAR(80)
decision: VARCHAR(20)
comment: TEXT
from_status: VARCHAR(20)
to_status: VARCHAR(20)
created_at: DATETIME
```

finding_id

evidence

```
PK id: VARCHAR(32)
run_id: VARCHAR(32)
finding_id: VARCHAR(32)?
type: VARCHAR(30)
title: VARCHAR(200)
payload: JSON
sha256: VARCHAR(64)
reproduce: TEXT
created_at: DATETIME
```

replays

```
PK id: INTEGER
FK experiment_id: VARCHAR(32)
kind: VARCHAR(15)
idx: INTEGER
seed: INTEGER
completed: BOOLEAN
passed: BOOLEAN?
judgement: JSON?
results: JSON
final_output: JSON
```